Security Project Proposal - A2 Trojan in the LLM era

Amber Tsao (cctsao), Jiahao Xu (e1583425)

Inspired by the brief survey of recent updates on how researchers tackle A2 trojans using LLMs, we are interested in gaining a deeper understanding of how these are done. We will mainly focus on two papers, including SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection (Chaudhuri et al., 2024) and LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies (Chaudhuri et al., 2025), which respectively represent the defending and attacking LLMs of the A2 trojans issue.

SPICED feedback was originally designed for designers to not only detect trojans but also identify weaknesses and improve the design. However, such feedback may also help the attacker side to hide the trojan even better. According to LATENT's report, the average evasion rate can be improved to 94.4% with feedback from 23.7% for one-shot prompting. We are curious about what role LLMs can play in this issue, and whether LLMs can be a solution to A2 Trojans or whether they mainly leak additional clues to the attacker.

To keep the scope feasible for a course project, we will reproduce the core components of both sides to form a minimal closed loop of the LLM-A2 trojan pipeline. We will replace HSPICE simulation with the open-source ngspice and look at 2 compact micro-benchmarks, a 2-stage op-amp and a 4-input AND gate. We will also skip SPICED's syntax-fixing evaluation to keep it simpler. If LLM-written netlists fail, we will limit their output to JSON parameters and add them to verified templates. We might also change the testing circuits if the detector is unreliable on the selected circuits.

We are planning to use ngspice, Python, and the OpenAI API with agent-assisted coding to work on two netlists, Trojan templates, and an automated attacker/detector script. Finally, we would do a small-scale analysis on how the level of detail of the feedback (i.e., none, binary, full) affects the evasion rate.

Reference

Yang, K., Hicks, M., Dong, Q., Austin, T., & Sylvester, D. (2016). A2: Analog Malicious Hardware. 2016 IEEE Symposium on Security and Privacy (SP), 18–37. https://doi.org/10.1109/SP.2016.10.

Jayeeta Chaudhuri, Dhruv Thapar, Arjun Chaudhuri, Farshad Firouzi, & Krishnendu Chakrabarty. (2024). SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection. https://arxiv.org/pdf/2408.16018.

Jayeeta Chaudhuri, Arjun Chaudhuri, & Krishnendu Chakrabarty. (2025). LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies. https://arxiv.org/html/2505.06364v1.